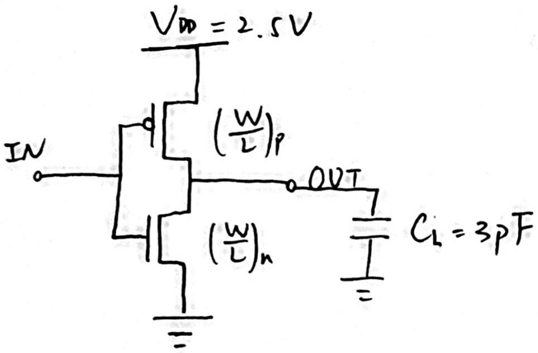
1.



	V_{TO}	γ	V_{DSAT}	k'	λ
NMOS	0.43	0.4	0.63	115×10^{-6}	0.06
PMOS	-0.4	-0.4	-1	-30×10^{-6}	-0.1

$$\cancel{t_{pLH} = \quad I_{DSATp} = k_p'\left(\frac{W}{L}\right)_p\left((V_{GS}-V_{THp})V_{DSAT} - 0.5V_{DSAT}^2\right)}$$

$$\cancel{R_{eq}} = \quad \cancel{t_{pLH}} = \quad \cancel{t_{pLH} = \frac{\frac{C_L V_{DD}}{2}}{I_{DSATp}}} =$$

$$\cancel{i = I_{DSATp}(1+\lambda_p V_{out}) = C\frac{dV_{out}}{dt}}$$

0V ~ 1.25V : $i = -I_{DSATp}\left(1+\lambda_p\left(V_{out}-V_{DD}\right)\right) = C_L\frac{dV_{out}}{dt}$

$$t_{pLH} = \frac{C_L}{-I_{DSATp}\lambda_p}\ln\left(\frac{1-\frac{1}{2}\lambda_p V_{DD}}{1-\lambda_p V_{DD}}\right) = 2nS \Rightarrow \left(\frac{W}{L}\right)_p = 3.83\times10^{1} \Leftrightarrow W_p = 9.58\mu m$$

2.5V ~ 1.25V $\cancel{t_{pHL}}$ =

$$I_{DSATn}(1+\lambda_n V_{out}) = -C_L\frac{dV_{out}}{dt}$$

$$t_{pHL} = \frac{C_L}{I_{DSATn}\lambda_n}\ln\left(\frac{1+\lambda_n V_{DD}}{1+\lambda_n\frac{V_{DD}}{2}}\right) = 2nS \Rightarrow \left(\frac{W}{L}\right)_n = 1.33\times10^{1} \Leftrightarrow W_n = 3.31\mu m$$

$$I_{DSAT} = k'\left(\frac{W}{L}\right)\left((V_{GS}-V_{TH})V_{DSAT} - 0.5V_{DSAT}^2\right)$$

$(V_{GS} = 0V)$ $\quad I_{DSATp} = -30\times10^{-6}\times\left(\frac{W}{L}\right)_p\times(-1)\times\left(-2.5+0.4-0.5\times(-1)\right)$

$$= -4.8\times10^{-5}\left(\frac{W}{L}\right)_p \ (A)$$

$(V_{GS} = 2.5V)$ $\quad I_{DSATn} = 115\times10^{-6}\times\left(\frac{W}{L}\right)_n\times0.63\times\left(2.5-0.43-0.5\times0.63\right)$

$$= 1.27\times10^{-4}\left(\frac{W}{L}\right)_n \ (A)$$

Spice 仿真：$t_{pHL} = 2.218ns$，$t_{pLH} = 1.829ns$

2. $f = \sqrt[3]{F} = \sqrt[3]{\frac{C_L}{C_{in,1}}} = 6.69$

$t_p = 3t_{p0}\left(1+\frac{\sqrt[3]{F}}{\gamma}\right) = 3\times 0.03\times(1+6.69) = 0.692ns$

第一级：$W_p = 1.2\mu m$，$W_n = 0.5\mu m$

第二级：$W_p = 8.03\mu m$，$W_n = 3.345\mu m$

第三级：$W_p = 53.7\mu m$，$W_n = 22.38\mu m$

Spice 仿真：$t_{pHL} = 689.7ps$，$t_{pLH} = 703.1ps$

$$t_p = \frac{t_{pHL}+t_{pLH}}{2} = 696.10ps$$

3. $\cancel{N=4,\ h = \sqrt[4]{H} = \sqrt[4]{\frac{C_L}{C_{in}}} = \sqrt{5}}$

$N=4$，$F = \frac{C_L}{C_{in}} = \frac{50C_{ref}}{2C_{ref}} = 25$，

$g_1 = g_4 = 1$，$g_2 = \frac{5}{3}$，$g_3 = \frac{3+2}{1+2} = \frac{5}{3}$

$G = \frac{25}{9}$

3. $N=4$，$F = \frac{C_L}{C_{in}} = \frac{50C_{ref}}{2C_{ref}} = 25$，$G = 1\times\frac{5}{3}\times\frac{5}{3}\times 1 = \frac{25}{9}$，$B = 2\times1\times1\times1 = 2$，$s_1 = 2$

$H = GBF = \frac{5^4\times 2}{9} \Rightarrow h = \sqrt[N]{H} = 5\times\sqrt[4]{\frac{2}{9}} = 3.433$

$f_1 = 3.433$，$f_2 = 2.060$，$f_3 = 2.060$，$f_4 = 3.433$

$s_2 = \left(\frac{f_1}{b_1}\right)\left(\frac{g_1}{g_2}\right)s_1 = \underset{a尺寸}{\underline{2.060}}$，$s_3 = \left(\frac{f_2}{b_2}\right)\left(\frac{f_1}{b_1}\right)s_1\left(\frac{g_1}{g_3}\right) = \underset{b尺寸}{\underline{4.243}}$，$s_4 = \left(\frac{f_3}{b_3}\right)\left(\frac{f_2}{b_2}\right)\left(\frac{f_1}{b_1}\right)s_1\left(\frac{g_1}{g_4}\right) = \underset{c尺寸}{\underline{24.278}}$

a：输入电容 $2.060C_{ref}$，与 C_{in} 的比 1.030

b：输入电容 $4.243C_{ref}$，与 C_{in} 的比 2.121

c：输入电容 $24.278C_{ref}$，与 C_{in} 之比：12.139.